

NETWORKING 03

TCP/IP MODEL



First Clear the Confusion

👉 TCP/IP model $\neq$ TCP and UDP

These are **different things**:

- **TCP/IP model** = full network structure (4 layers)
- **TCP & UDP** = protocols inside **Transport layer**
- 👉 TCP/IP model is like a **company**
- Application → Boss (apps)
- Transport → Workers (TCP/UDP)
- Internet → Delivery route
- Network → Roads

👉 So **TCP & UDP** are workers inside Transport layer



TCP/IP MODEL (Practical Model)

👉 TCP/IP model consists of **4 layers** and is used in real-world networking.



Mapping with OSI Model

TCP/IP Layer	OSI Layers Covered
Application	7, 6, 5
Transport	4
Internet	3
Network Access	2, 1



Layers Explanation

1. Application Layer

👉 Combines OSI layers 5, 6, 7

👉 Functions:

- User interaction
- Data formatting
- Encryption 🔒
- Session management

👉 Protocols:

- HTTP, HTTPS
- FTP
- SSL/TLS

👉 Important:

👉 Work is done by protocols, not the layer itself.

2. Transport Layer

👉 Responsible for data transmission

👉 Protocols:

- TCP → reliable
- UDP → fast but unreliable

👉 Functions:

- Error checking
- Flow control

3. Internet Layer

👉 Responsible for routing

👉 Uses IP addressing

4. Network Access Layer

👉 Physical transmission of data

👉 Includes:

- Cables
- WiFi
- Hardware

OSI Layer No	OSI Layer Name	Work Done (OSI)	TCP/IP Layer	Work Done (TCP/IP)
7	Application	User interface, network services	Application	Same work + protocols handle it
6	Presentation	Encryption, compression, formatting	Application	Done by TLS/SSL, HTTP etc
5	Session	Session management (start/stop connection)	Application	Managed by protocols
4	Transport	Reliable data transfer, segmentation	Transport	TCP (reliable), UDP (fast)
3	Network	Routing, IP addressing	Internet	IP-based routing
2	Data Link	Frame transfer, MAC addressing	Network Access	Hardware-level transfer
1	Physical	Transmission of bits (cables, signals)	Network Access	Same physical transmission

- 👉 **OSI = 7 detailed layers**
- 👉 **TCP/IP = 4 practical layers**

👉 TCP/IP merges:

- (7+6+5) → Application
- (2+1) → Network Access

OSI Model (7 Layers)

1. Physical → sends raw bits
2. Data Link → MAC, frames
3. Network → IP, routing
4. Transport → delivery (TCP/UDP)
5. Session → connection control
6. Presentation → encryption
7. Application → user interface

TCP/IP Model (4 Layers)

1. Network Access → hardware + transmission
2. Internet → routing (IP)
3. Transport → TCP/UDP
4. Application → user + encryption + session